

qRAT

Software Documentation & User Guide

Version 0.7.0

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Contents

1 About **3**

2 How to run qRAT? **3**

3 Application Layout **4**

4 General Data Input **4**

5 Data Format **5**

6 Supported Instruments **7**

7 How to analyse data **8**

8 Results (Tables and Plots) **10**

9 Data Quality and Filtering **12**
9.1 Automatic Filtering 12
9.2 Manual Filtering and Controls 12

10 Reference Gene Selection (Reference Finder) **13**

11 Computation of Relative Quantities **13**
11.1 Mathematical Framework 14
11.2 Normalization with Multiple Targets 14
11.3 Reference Gene Stability 14

12 Biological Replicates and Grouping **14**

13 Inter-Plate Calibration **15**
13.1 Experimental Strategies 15
13.2 Implementation in qRAT 16

14 Statistics **16**
14.1 The Moderated T-Statistic 17
14.2 Normalization for the Statistics 17
14.3 Multiple Testing and P-Value Adjustment 17
14.4 Group Comparisons 17

15 R Packages and Third-Party Software **19**

16 Release Notes **20**

1 About

qRAT is an R-based application to automate the processing of raw Quantification Cycle (Cq) data files exported from virtually any qPCR instrument using well established and state-of-the-art statistical and graphical techniques. The purpose of this tool is to provide a comprehensive, straightforward, and easy-to-use solution for the **relative quantification** of RT-qPCR data that requires no programming knowledge or additional software installation.

The current implementation allows **ΔCq** calculation (relative to endogenous control(s)), **ΔΔCq** calculation (relative to endogenous control(s) and a reference sample), **inter-plate variation correction**, and automated **reference gene stability analysis**. Moreover, functionalities for **parsing**, **filtering** and **visualisation** of relative RT-qPCR data are included.

2 How to run qRAT?

Since version 0.5, qRAT is no longer available as a standalone installer. You can access the application in two ways:

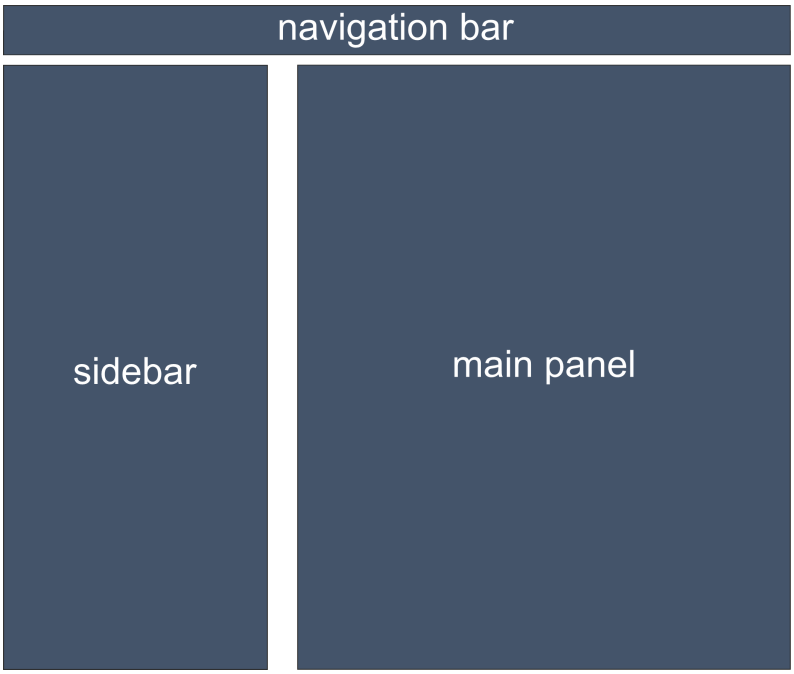
- **Web Application:** Use the hosted version at **qrat.shinyapps.io/qrat/**.
- **Local Installation (via R):** Follow the instructions on the **qRAT GitHub repository**.

To test the application, you can download a **Sample Dataset here** (single files and a description: **example_data** folder on GitHub). Additional information and updates can be found on the **qRAT Homepage**.

3 Application Layout

The main window of qRAT consists of 3 components:

1. **Top navigation bar:** Browse between Single Plate, Multiple Plates, and Help sections.
2. **Sidebar:** Dynamic settings for data input, filtering, and plot customization.
3. **Main panel:** Interactive tabs containing tables and plots.



4 General Data Input

qRAT processes **quantification cycle (Cq) values** measured by a qPCR instrument and exported by that instrument's software. Depending on the instrument software that generates the data, Cq values may have alternative names such as Ct, Cp, or TOP. Cq is the official abbreviation in the **Real-time PCR Data Markup Language** which is considered as the universal data standard for exchanging quantitative PCR (qPCR) data. To start any qRAT execution, users must import **txt or csv** files containing Cq values for every well. Files can be **tab-, comma-, or semicolon**-delimited. The data import is made to be flexible and is capable of reading export files generated by most qPCR instrument software programs due to its ability to

recognize synonymous column names. The import procedure does not alter the original data file.

A list of **supported qPCR instruments** and software programs is included in section *Supported Instruments*. However, if your instrument or software is **not listed** or you experience **problems importing the data** into qRAT, please try renaming the columns in your exported file to **match the format** you see in section *Data Format*.

5 Data Format

Some qPCR machines add a header in the first rows of the file, qRAT automatically discards these header rows before parsing the content of the files. For this reason, the first column of the columns row must start by the word **"Well"**. Further requirements are that the columns **"Sample"**, **"Gene"** and **"Cq"** are present. Common synonyms are recognized automatically (case-insensitive): *Sample, Sample Name, Name* for the sample, *Gene, Target, Target Name, Detector, Primer/Probe* for the gene and *Cq, Ct* for the Cq value. If a file contains several matching columns (e.g. "Sample" and "Sample Name"), the first one in this order is used. Sample and gene names may contain spaces and special characters, and sample names may start with a digit (e.g. *24h*). Technical replicates of the same biological sample must have **exactly the same name**; the number of technical replicates should ideally be identical across all samples. Empty wells/blank rows (without sample name or gene target) are automatically removed. If a required column is missing, qRAT reports it in the validation dialog and does not start the analysis. This is represented in the table below:

Well	Sample	Gene	Cq
A1	Sample1	Target1	22.2
A2	Sample1	Target1	21.6
A3	Sample1	Target1	21.2
B1	Sample1	Target2	33.3
B2	Sample1	Target2	29.9
B3	Sample1	Target2	30.1
C1	Sample2	Target1	24.1
C2	Sample2	Target1	23.8
C3	Sample2	Target1	24.3
D1	Sample2	Target2	26.2
D2	Sample2	Target2	26.6
D3	Sample2	Target2	26.4

In case your samples are spread across multiple plates, you may want to make

use of inter-plate calibration. For this, the **calibrator must have exactly the same name** on all plates:

Well	Sample	Gene	Cq
A1	Sample1	Target1	22.2
A2	Sample1	Target1	21.6
A3	Sample1	Target1	21.2
B1	Sample1	Target2	33.3
B2	Sample1	Target2	29.9
B3	Sample1	Target2	30.1
C1	Sample2	Target1	24.1
C2	Sample2	Target1	23.8
C3	Sample2	Target1	24.3
D1	Sample2	Target2	26.2
D2	Sample2	Target2	26.6
D3	Sample2	Target2	26.4
E1	IPC	Target3	23.6
E2	IPC	Target3	23.4
E3	IPC	Target3	23.6

6 Supported Instruments

For most users there is no need to modify the data file as qRAT supports **exported files** from the majority of the real-time PCR instruments and accompanying data analysis software as long as they are **tab-, comma-, or semicolon**-delimited **csv or txt** files. See section *General Data Input* for more details. If your instrument or analysis software is not supported, you are advised to **modify your exported txt or csv file** so it matches the required data format as described in section *Data Format*.

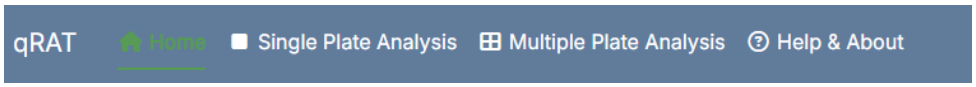
Technical replicates of the same biological sample must have **exactly the same name**. Empty wells/blank rows (without sample name or gene target) are automatically removed. For *Multiple Plates Analysis* the **inter-plate calibrator** must have exactly the **same name** on all plates; a plate file without the required columns is excluded from the analysis and reported in a notification.

Instruments/Software: qRAT was successfully tested with *txt/csv* files from Analytik Jena (qTower 3G), Wafergene (SmartCycler), Applied Biosystems (5700, 7000, 7300, 7500, 7900 SDS 2.4, StepOne(Plus), Viia7).

7 How to analyse data

This section outlines the standard workflow for performing a relative quantification analysis in qRAT. The application is designed to guide the user from raw data import to statistical validation.

1. **Plate Setup Selection:** Begin by choosing your experimental setup in the top navigation bar: **Single Plate** for data within one file, or **Multiple Plates** for experiments spanning several runs.



2. **Sidebar Navigation (Inputs):** Use the **sidebar** (Data Input, Analysis Input) to configure your analysis. It is recommended to proceed from top to bottom:
   (a) **Data Input:** Select your plate mode.
   (b) **Formatting:** Choose the delimiter of your file (Auto, Comma, Tab, Semicolon). "Auto" is generally recommended as it detects the format automatically.
   (c) **Upload:** Upload your file(s). In "Multiple Plates" mode, select all relevant files simultaneously (e.g., by holding the [Ctrl] or [Cmd] key). The **main panel** will update as soon as the upload is complete.
3. **Analysis Workflow (Main Panel):** Navigate through the tabs to refine your analysis. All calculations are reactive and update instantly.
   (a) **Raw Data:** Provides an initial overview. The spatial layout plot helps identify plate-wide patterns or pipetting artifacts before filtering.
   (b) **Reference Finder:** If you are unsure which endogenous controls are the most stable, use this tab. It employs the **GeNorm algorithm** to rank your genes. Stable genes identified here should be selected in the subsequent analysis steps.
   (c) **Filtering and Quality:** Define your quality criteria. You can toggle between **Automatic Filtering** (SD and Cq cut-offs) and **Manual Filtering**. Individual wells can be excluded as outliers in the *Select Outliers* dialog or by clicking them in the *Plate View*.

(d) **Grouping (Biological Replicates):** If your samples represent biological replicates, use this tab to assign them to common group names (e.g., "WT" or "Mutant"). This is crucial for valid statistical testing based on biological variance.

4. **Relative Expression Analysis:** Once the data is cleaned and grouped, proceed to quantification:

   (a) **Relative dCq (ΔCq):** Select your reference gene(s). The main panel displays the ΔCq values and Relative Quantities ($RQ = 2^{-\Delta Cq}$).

   (b) **Relative ddCq (ΔΔCq):** Select both reference gene(s) and a **calibrator sample** (control). This normalizes the data relative to the control group, providing Fold Change ($FC = 2^{-\Delta\Delta Cq}$) results.

5. **Statistical Validation:** qRAT utilizes the *limma* package for robust statistical analysis. You can compare two or multiple groups.

   (a) **Comparison Type:** Choose between a *Single Comparison* (all samples against one control sample) and *Multiple paired Comparisons* (freely defined pairs of samples, e.g. "WT T1 vs WT C"). All comparisons use the moderated t-statistic.

   (b) **P-Value Adjustment:** To control for multiple testing, select an adjustment method (Benjamini-Hochberg (default), Holm, or Bonferroni).

   (c) **Significance Levels:** Results are presented with t-statistics, p-values, and significance asterisks.

Table 1: Significance levels used in qRAT (based on the adjusted p-value; identical in the statistics tables and the plots).

Symbol	P-Value Range
ns	$p > 0.05$
*	$p \leq 0.05$
**	$p \leq 0.01$
***	$p \leq 0.001$
****	$p \leq 0.0001$

8 Results (Tables and Plots)

qRAT is built on a reactive framework, meaning it recalculates all intermediate and final results (quality controls, normalization, and statistics) instantaneously. As soon as a parameter in the sidebar is modified, all tables and plots will **automatically update** to reflect the current analysis settings.

Tables: All data tables are equipped with an interactive header providing **data export** functions. These allow users to copy data to the clipboard or export it into various formats such as CSV, Excel, or PDF, making it easy to include raw or processed results as supplemental data in publications. Result tables for ΔCq and ΔΔCq analysis provide a comprehensive breakdown of every gene/sample combination. This includes calculated ΔCq/ΔΔCq values, their standard deviation, the number of valid technical replicates of the target gene (*n*), and the resulting relative quantification (RQ) or fold change (FC). To ensure readability, values in the tables are rounded to two decimal places. If the target or the reference gene has fewer than two valid replicates in a sample, no standard deviation can be computed: the value is still shown (without SD and without error bar) and a notification lists the affected sample/gene combinations.

Plots: All visualizations in qRAT are fully customizable and designed for publication-ready output. It is important to note that changes in the appearance settings **only affect the visual representation**, while the underlying statistical calculations remain untouched.

By default, qRAT includes all genes and samples in the plots. However, users can precisely control the content:

- **Gene Filtering:** Select or deselect specific target genes via the "Filter Genes" drop-down menu to focus on relevant results.
- **Sample Selection:** Include or exclude specific samples or groups to simplify the comparison.
- **Sample Ordering:** Using the *Sample Ordering* feature in the *Plot Appearance* section of the sidebar, users can reorder the samples on the x-axis via drag-and-drop to match the experimental design.

The plot settings in the sidebar are grouped into collapsible sections: *Data Selection*, *Data Visualization*, *Plot Appearance* and *Export*. Plots are interactive: hovering over a bar or point shows the underlying values (including the error and *n*), the toolbar in the top right corner of a plot offers zoom and a PNG snapshot, and

every card can be expanded to full screen. For permanent export, qRAT provides an export method:

1. **Advanced Export:** The dedicated *Export* section in the sidebar allows for precise control over the output. Users can specify the **file format** (PDF, SVG, PNG), the **image dimensions** (width and height in inches), and the **resolution scale**. This ensures that exported graphics meet the strict requirements of scientific journals.

Plot Customization (Appearance): Additional parameters allow for deep customization of the charts:

- **Plot Data & Type:** Choose between plotting ΔCq/ΔΔCq or RQ/FC values, and toggle between bar charts and dot plots.
- **Axes & Labels:** Users can define custom titles for the plot, the legend, and both axes. To accommodate long sample names, the **rotation angle** of the x-axis labels and the overall **font size** can be adjusted via sliders.
- **Error Bars & Scales:** Toggle the display of error bars (SD or SEM; the SEM is calculated as $\mathrm{SD}/\sqrt{n}$ with the number of valid replicates n) and switch the y-axis scale between linear, log2, or log10.
- **Significance:** Optionally show the significance asterisks of the statistics (Table 1) above the bars.
- **Colors:** A variety of professional color palettes from the **RColorBrewer** package are available to distinguish between samples or groups effectively.

9 Data Quality and Filtering

Quality control is a cornerstone of robust qPCR analysis. In the *Filtering and Quality* tab, qRAT combines automated algorithms with manual oversight to ensure data integrity. All excluded data points are summarized in a dedicated table and are omitted from all downstream calculations.

9.1 Automatic Filtering

Automated filtering criteria are based on two main parameters: replicate consistency and absolute Cq boundaries.

- **Replicate Variability:** qRAT identifies technical replicates by matching identical sample and gene names. To ensure precision, the standard deviation (SD) of these replicates is calculated. If the SD exceeds the user-defined threshold (default: 0.3, as per **MIQE Guidelines**), the outlier furthest from the mean is automatically removed. This is repeated as long as the SD still exceeds the threshold, but at least two replicates are always kept: with two discordant values it cannot be decided which one is the outlier, so both remain (and their high SD is visible in the results). The Cq cut-off below is applied first, so that values outside the valid range do not distort the SD.
- **Cq Cut-off:** This parameter defines the biological and technical detection limits. Values outside the specified range (default: 5–35) are treated as *NA*. This is essential for removing background noise or late-cycle artifacts.

9.2 Manual Filtering and Controls

Beyond automated rules, qRAT provides full control over the dataset through interactive elements.

- **Manual Outlier Selection:** Switch to *Manual Filtering* in the sidebar and open the *Select Outliers* dialog to exclude specific wells by clicking on their rows. Alternatively, click the wells directly in the *Plate View* (Raw Data tab) and apply them with *Exclude selected wells*; in Multiple Plates mode this applies to the plate shown, exclusions on other plates are kept. This is intended for removing known technical artifacts (e.g., air bubbles or evaporation) that might not trigger the automated filter.

- **Experimental Controls:** Users can designate specific samples as Non-Template Controls (NTC) or Reverse Transcriptase minus (RT-) controls. Selecting these samples in the sidebar excludes them from the relative quantification process to prevent them from biasing the experimental groups.

According to **MIQE Guidelines**, the number of technical replicates should be $n \geq 2$. If filtering reduces the number of replicates for a sample below this limit, qRAT will display a warning message in the main panel to alert the user to the reduced statistical power.

10 Reference Gene Selection (Reference Finder)

The stability of reference genes is critical for accurate normalization. qRAT now includes an automated **Reference Finder** tool.

In the *Reference Finder* tab, qRAT implements the **GeNorm** algorithm to rank candidate reference genes based on their expression stability (M-value).

- **M-Value:** Measures the average pairwise variation of a gene against all other candidate genes.
- **Selection:** Lower M-values indicate higher stability. Genes with $M < 0.5$ in homogeneous samples are generally considered excellent.
- **Ranking:** The least stable gene is excluded step by step and M is recalculated for the remaining genes, until the two most stable genes remain; they share rank 1. *Apply Top Genes as Refs* selects this pair as reference genes.
- **Input:** The calculation uses the mean Cq per sample and gene (after filtering, and after inter-plate calibration if activated). geNorm compares the genes across all samples, so only samples with a value for every candidate gene are used; at least two candidate genes and two samples are required.

Use the generated stability plot to identify the most robust targets before proceeding to ΔCq or ΔΔCq analysis.

11 Computation of Relative Quantities

The core of qRAT's analysis engine is the **ddCt R package**. It utilizes an optimized ΔΔCq method that assumes high and comparable amplification efficiencies across all targets, eliminating the need for a standard curve for every run.

11.1 Mathematical Framework

For the ΔΔCq method to be statistically valid, the amplification efficiencies of the target and reference primers should be approximately equal and close to 2 (representing 100 % efficiency). Relative quantification in qRAT is expressed in two ways:

- **Relative Quantity (RQ):** Used in ΔCq analysis, where samples are normalized to endogenous controls:

$$RQ = 2^{-\Delta Cq} \tag{1}$$

- **Fold Change (FC):** Used in ΔΔCq analysis, where samples are normalized to endogenous controls and expressed relative to a calibrator sample:

$$FC = 2^{-\Delta\Delta Cq} \tag{2}$$

11.2 Normalization with Multiple Targets

To increase the robustness of the results, qRAT supports the use of multiple reference genes and calibrator samples:

- **Reference Genes:** If multiple housekeeping genes are selected, the algorithm calculates the **arithmetic mean of their Cq values** for normalization.
- **Calibrator Sample:** One sample is selected as calibrator (control); its ΔCq values form the baseline, i.e. its fold change is 1 (reference line).

11.3 Reference Gene Stability

The accuracy of the relative quantification depends entirely on the stability of the chosen reference genes. While 2–4 reference genes are generally recommended, their stability must be experimentally verified.

qRAT now features an integrated **Reference Finder** tab based on the **geNorm** algorithm. It is highly recommended to use this tool to rank your candidate genes by their M-value stability score before proceeding with the final quantification.

12 Biological Replicates and Grouping

In many experiments, multiple biological samples receive the same treatment. To analyze these correctly as one experimental group, qRAT provides a specialized

Grouping tab. This allows users to aggregate individual biological replicates into a single group for statistical comparison and plotting.

Example Assignment: Consider an experiment with three control animals and three treated animals. In the input file, each biological sample must have a unique name. In the qRAT Grouping tab, they are assigned as follows:

Table 2: Biological Grouping Example

Sample Name (Input)	Assigned Group (Grouping Tab)	Role
Sample_1	Control	Biological Replicate 1
Sample_2	Control	Biological Replicate 2
Sample_3	Control	Biological Replicate 3
Sample_4	Treatment_A	Biological Replicate 1
Sample_5	Treatment_A	Biological Replicate 2
Sample_6	Treatment_A	Biological Replicate 3

By defining these groups, qRAT will:

- **Aggregate Data:** First average the technical replicates of every biological sample, so each biological replicate enters its group with one value; then calculate the mean and standard deviation/error based on the $n = 3$ biological replicates per group (column *n* in the result tables). Samples that are not assigned to a group are treated as a group of their own with $n = 1$ (no standard deviation). The quality plots in the *Filtering and Quality* tab keep showing the individual technical replicates.
- **Statistical Power:** Perform moderated t-statistics based on the variance between the groups rather than between individual technical replicates.
- **Visualisation:** Plots will display the results for "Control" and "Treatment_A" as single entities.

13 Inter-Plate Calibration

Inter-plate variation is a common but often underestimated source of error in qPCR experiments spanning multiple runs. qRAT provides a dedicated **Inter-Plate Calibration (IPC)** procedure to detect and remove these technical offsets.

13.1 Experimental Strategies

To minimize inter-plate variation, two main strategies are generally recognized:

- **Sample Maximization:** This is the preferred strategy. All samples for a specific gene are analyzed within the same run/plate. Since all comparisons for that gene occur under identical conditions, no inter-plate calibration is required for the samples. In this setup, reference genes do not necessarily need to be on the same plate as the target gene, as they serve to quantify relative concentrations between samples.
- **Inter-Plate Calibrators (IPC):** If sample maximization is not feasible (e.g., due to large sample sizes), a small set of identical samples—the IPCs—must be repeated across all plates. These calibrators allow qRAT to calculate a correction factor to harmonize the data between different runs.

13.2 Implementation in qRAT

To activate the correction, navigate to the **Inter-Plate Calibration** tab, switch on *Use Calibration* in the sidebar and select the **IPC sample**. The IPC sample itself is then removed from the analysed samples.

For each plate, qRAT calculates the mean Cq of the IPC on that plate and its deviation from the mean IPC Cq over all plates (the calibration factor, shown in the *Calibration Factors* table). This offset is subtracted from all Cq values of the plate:

$$Cq_{\text{calibrated}} = Cq_{\text{raw}} - (\overline{Cq}_{\text{IPC, plate}} - \overline{Cq}_{\text{IPC, all plates}}) \quad (3)$$

The calibrated Cq values are used for all downstream results: ΔCq, ΔΔCq, the statistics and the Reference Finder. The *Calibration Check Plot* compares raw and calibrated values. Plates without IPC values cannot be calibrated; their Cq values are set to NA and a warning is shown.

The Inter-Plate Calibration function requires that the IPC samples have **exactly the same name** across all plates. It is only valid if the identical physical material (same cDNA/DNA source) is used as a calibrator on every plate being compared.

14 Statistics

qRAT employs a sophisticated statistical framework to evaluate the significance of differences in gene expression between samples or experimental groups. All calculations are powered by the **limma package**, which is specifically designed for high-throughput genomic data.

14.1 The Moderated T-Statistic

The core of the analysis is the **moderated t-statistic**. While it is interpreted similarly to a standard Student's t-test, it is much more robust for experiments with small sample sizes. Using a **Bayesian model**, the standard errors are "moderated" across all genes—essentially "squeezing" them towards a common global value. This approach reduces the impact of chance outliers and provides more reliable results when the number of replicates is limited.

14.2 Normalization for the Statistics

For the statistical tests, every technical replicate of a target gene is normalized against the reference gene(s) of the same sample:

$$\Delta Cq_{\text{replicate}} = Cq_{\text{target, replicate}} - \overline{Cq}_{\text{reference, sample}} \tag{4}$$

where $\overline{Cq}_{\text{reference, sample}}$ is the mean of all valid reference gene Cq values of that sample. This is the same basis as in the ΔCq/ΔΔCq calculation and keeps all valid target replicates, even if individual reference replicates were removed by filtering.

14.3 Multiple Testing and P-Value Adjustment

When comparing multiple genes or groups simultaneously, the probability of encountering false positives increases. To maintain statistical integrity, qRAT provides several methods to adjust the p-values:

- **Benjamini & Hochberg (FDR):** The default method. It controls the False Discovery Rate, balancing the power to find true differences with the risk of false positives.
- **Bonferroni:** A more conservative approach that strictly controls the family-wise error rate by dividing the alpha level by the number of tests.
- **Holm:** A step-down procedure that is generally more powerful than the Bonferroni method while maintaining similar strictness.

14.4 Group Comparisons

Users can compare **all samples against one control** (*Single Comparison*) or define **pairs of samples** (*Multiple paired Comparisons*, labelled e.g. "WT T1 vs WT C").

At least two samples are required; otherwise the statistics table shows a hint instead of results. The results in the main panel include:

- **p.Value:** The raw, unadjusted p-value.
- **adj.P.Value:** The p-value after applying the selected adjustment method.
- **Significance:** Asterisks indicating the level of statistical significance based on the adjusted p-value.

For valid statistical results, ensure that biological replicates have been correctly assigned in the **Grouping** tab, as this ensures the variance is calculated based on biological rather than technical replicates.

15 R Packages and Third-Party Software

qRAT is built using the R statistical programming language and the Shiny web framework. All core qPCR calculations and visualizations rely on high-quality, peer-reviewed Bioconductor and CRAN packages.

Table 3: Software and R Packages

Category	Software / Package
Core Environment	R, RStudio
Framework & UI	Shiny, bslib, shinyjs, shinyWidgets, shinyjqui, waiter, DT, bsicons
qPCR Analysis	ddCt, HTqPCR, limma
Data Processing	dplyr, data.table, rlang, tidyr
Visualization	ggplot2, ggiraph, ggpubr, scales, RColorBrewer, svglite
Reference Finder	ctrlGene (geNorm algorithm)

Please note that these packages may have additional dependencies. For detailed documentation on a specific tool, please refer to its respective CRAN or Bioconductor project page.

Icons used in the application are provided by *Font Awesome*, *Bootstrap Icons* and *Freepik* (via www.flaticon.com). The font *Inter* (SIL Open Font License) is bundled with the application.

16 Release Notes

- **qRAT 0.7.0 (Current)**

  Note: Results can differ from version 0.6.0 (replicate filter, p-value adjustment, inter-plate calibration, biological grouping, normalization for the statistics, Reference Finder). Please re-run analyses that were done with 0.6.0 before reporting them.

  - **Fixed:** The selected p-value adjustment is applied. In 0.6.0 Bonferroni was always used, regardless of the selection.
  - **Fixed:** Inter-plate calibration is applied to ΔCq, ΔΔCq, the statistics and the Reference Finder. In 0.6.0 the calibrated values were only displayed.
  - **Fixed:** Multiple paired comparisons react to the selected sample pairs (in 0.6.0 no results were shown).
  - **Fixed:** Values without standard deviation (fewer than two valid replicates) are shown instead of being silently removed; the SEM uses the real number of replicates.
  - **Fixed:** Faulty uploads (missing columns, wrong file type) no longer end the session; sample names may start with a digit.
  - **Fixed:** Several settings of the Multiple Plates plots (legend position, plot export) used the Single Plate settings.
  - **Fixed:** The automatic replicate filter works as documented (SD above the threshold: remove the replicate furthest from the mean, keep at least two). In 0.6.0 all values further than the threshold from the mean were removed, which could discard all replicates of a sample because of a single outlier.
  - **Fixed:** The Reference Finder uses geNorm (*M* value) as documented. In 0.6.0 a simplified stability measure was calculated and labelled as NormFinder.
  - **Fixed:** Biological grouping averages the technical replicates of each sample first, so mean, SD/SEM and statistics are based on the biological replicates. In 0.6.0 all technical replicates of a group were pooled.
  - **Fixed:** The statistics no longer fail when a sample has only one value (e.g. a single biological replicate).

  - **Changed:** Statistics normalize every target replicate against the mean reference Cq of its sample; no more missing results when single reference replicates were filtered.
  - **Changed:** Unified significance symbols in tables and plots (ns, *, **, ***, ****).
  - **Added:** Exclusion of wells by clicking them in the Plate View; column *n* (valid replicates) in the result tables.
  - **Improved:** Interactive plots with ggiraph (identical to the exported figures), collapsible sidebar sections, full-screen plots, faster and more robust data handling.

- **qRAT 0.6.0**
  - **Added: Reference Finder** module using the geNorm algorithm (*M*-value stability ranking).
  - **Added:** Manual Filtering via interactive well selection.
  - **Added:** Biological Grouping functionality for replicate aggregation.
  - **Improved:** Full UI migration to Bootstrap 5 using the *bslib* library.
  - **Improved:** Optimized reactivity for data analysis.
  - **Improved:** Enhanced plot customization (adjustable font sizes, axis titles, and label rotation).

- **qRAT 0.1.0 – 0.5.0**
  - Initial releases: basic ΔΔCq calculation, CSV import, and fundamental plotting.